

Lab 4

***Submit as a Microsoft Word document in Dropbox area of D2L titled Lab 4 submission ***

Functional Dependencies

Determine functional dependencies of the follow tables. When stating your answers use standard notation for showing functional dependencies. Example:

SKU $\rightarrow$ SKU_Description, Department, Buyer

1. Consider the table:

STAFF_MEETING (EmployeeName, ProjectName, Date)

The rows of this table record the fact that an employee from a particular project attended a meeting on the given date. Assume that a project meets at most once per day. Also, assume that only one employee represents a given project, but that employees can be assigned to multiple projects. State the functional dependencies **only**.

ProjectName, Date $\rightarrow$ EmployeeName

ProjectName $\rightarrow$ EmployeeName

2. Consider the table:

STUDENT (Number, Name, Dorm, RoomType, DormCost, Club, ClubCost, Sibling, Nickname)

Assume that students pay different dorm costs, depending on the type of room they have, but that all members of a club pay the same cost. Assume that students can have multiple nicknames. State the functional dependencies **only**.

Number $\rightarrow$ Name, RoomType, Club, Dorm, Sibling

RoomType $\rightarrow$ DormCost

Normalization

3. Normalize the following table into a set of tables that puts them into 3NF.

Write your answer in the standard format:

Example of Standard format - STUDENT(Stid,Sname,Classid)

The data provided below is in one table. Put the data in this table into 3rd normal form (3NF). [Note: Eid = Employee id, Pid = project id, Mid = Manager id]

Eid	Ename	Pid	Pname	Mid	Mname
E01	John	p1	Eco DBMS	m1	Becky
E02	Mary	p2	Admin	m2	Joe
E03	Mark	p2	Admin	m2	Joe
E04	Travis	p3	Educ.	m1	Becky

a) Identify functional dependencies.

Eid-> Ename

Pid-> Pname,Mid

Mid-> Mname

b) Normalize the listing into 3NF.

Employee(Eid,Ename)

Project(Pid,Pname,Mname)

Manager(Mid,Mname)

ProjectEmployees(Eid,Pid)

4. Normalize the Veterinary Office List shown in the figure below.

	A	B	C	D	E	F	G	H	I	J
1	PetName	Type	Breed	DOB	Owner	OwnerPhone	OwnerEmail	Service	Date	Charge
2	Kino	Dog	Std. Poodle	27-Feb-05	Marsha Downs	555-123-6788	MD@somewhere.com	Ear Infection	17-Aug-06	\$ 65.00
3	Teddy	Cat	Cashmier	1-Feb-04	Richard James	555-444-0098	RJ@somewhere.com	Nail Clip	5-Sep-06	\$ 27.50
4	Filo	Dog	Std. Poodle	17-Jul-06	Marsha Downs	555-123-6788	MD@somewhere.com			
5	AJ	Dog	Collie Mix	5-May-05	Liz Frier	555-444-5596	LF@somewhere.com	One year shots	5-May-06	\$ 42.50
6	Cedro	Cat	Unknown	6-Jun-01	Richard James	555-444-0098	RJ@somewhere.com	Nail Clip	5-Sep-06	\$ 27.50
7	Woolley	Cat	Unknown	???	Richard James	555-444-0098	RJ@somewhere.com	Skin Infection	3-Oct-06	\$ 35.00
8	Buster	Dog	Border Collie	11-Dec-00	Miles Trent	555-999-8861	MT@somewhere.com	Laceration Repair	5-Oct-06	\$ 127.00

a) Identify functional dependencies.

Owner->OwnerPhone,OwnerEmail

Service->Date

PetName-> Type,Breed,DOB,Owner

Club -> ClubCost

b) Normalize the listing into 3NF.

Owner(Owner, OwnerPhone, OwnerEmail)

Service(Service, Date)

Pet(PetName, Type, Breed, DOB, Owner)

Appointment(PetName, Service, Date)

5. **Normalization Matching. Indicate which normal form, each of the following conditions represents. [1NF, 2NF, 3NF, 4NF]**

	Normal Form
1) No Partial Dependencies	2NF
2) No multi-value attributes	1NF
3) No dependency between non-key attributes	3NF
4) All attributes are dependent on the key	2NF
5) All attributes are dependent on the whole key	2NF
6) All attributes are dependent on nothing but the key	3NF
7) Similar to BCNF	3NF
8) No transitive dependencies	3NF
9) For a table to be considered a relation, it must at least satisfy	1NF

6. **For the following, answer whether Tables are in 3NF?**

If No, State the normalization rule that is violated. Rules can be:

A) No multi-valued attribute

B) Not dependent on the whole primary key

C) Dependencies between non-key attributes.

7. Employee (ssn, Name, Salary, Address, ListOfSkills)

Yes **No** Violates Rule: A

8. Vehicle (LicensePlate, Brand, Model, PurchasePrice, Year, OwnerSSN, OwnerName)

Yes **No** Violates Rule: C

9. Employee (ssn, Name, Salary, did)

Yes No Violates Rule:

10. Customer (Cust_Id, Name, Salesperson, Region) where Salesperson determines Region.

Yes **No** Violates Rule: C

11. Component (ItemNo, ComponentNo, ItemName, Quantity) where ItemNo -> ItemName

Yes **No** Violates Rule:

